

# Data Science & AI/ML

## Student Handout – Lecture 1: Welcome to the Course

80-Hour Data Science & Machine Learning Program | 7 Modules

Welcome! Over the next 80 hours, you will go from writing your first line of Python to building a system that classifies plant diseases from photos and one that forecasts sales for an e-commerce business. This handout explains, in plain language, what Data Science actually is, how AI/ML/DL relate to each other, and what the road ahead looks like.

### 1 What is Data Science?

**In one line:** Data Science is the practice of turning raw, messy data into useful answers – using statistics, code, and knowledge of the subject at hand.

It sits at the intersection of three skills. You don't need to be a master of all three on day one, but by the end of this course you'll have working competence in each:

- **Statistics** – to reason correctly under uncertainty (Module 2).
- **Programming** – to actually process data at scale (Module 1, and throughout).
- **Domain knowledge** – to ask the right question and know if the answer makes sense.

#### Delivery Time Prediction

When Zomato or Swiggy tells you “Delivery in 28 minutes,” that's not a guess – it's a model's prediction, trained on thousands of past deliveries using inputs like distance, restaurant prep time, current traffic, and rider availability. Each of these inputs is called a **feature**, and you'll learn to build exactly this kind of predictor in Module 3.

#### Recommendations & Sports Analytics

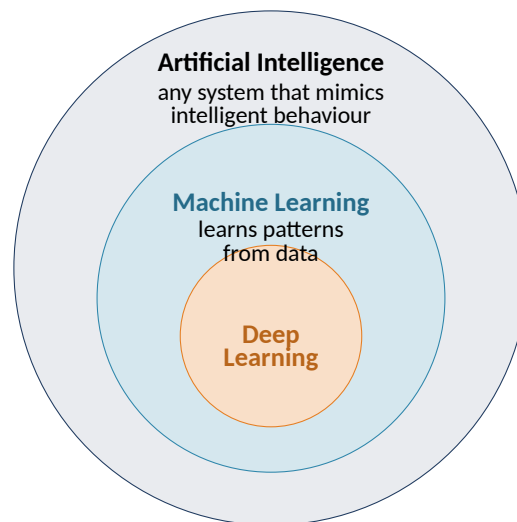
Netflix's “Because you watched...” row compares your viewing pattern to millions of similar users. During an IPL match, the live “win probability” number is a model estimating outcome likelihood from the current score, wickets, and overs remaining – learned from thousands of past matches. You will work with an IPL-style dataset in your very first mini-project.

#### Key Takeaway

A useful reality check for the field: roughly **80% of real Data Science work** is collecting and cleaning data – not building fancy models. Expect labs in this course to reflect that honestly.

### 2 AI vs Machine Learning vs Deep Learning

These three terms are often used interchangeably in the media, but they are *nested* – like three circles, one inside the other.



| Layer     | What it means                                                                | Simple test                                                                           |
|-----------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>AI</b> | Any system built to act intelligently, whether or not it "learns"            | Does it just follow fixed rules a human wrote, or does it improve from experience?    |
| <b>ML</b> | A subset of AI where the system <i>learns the rules itself</i> from examples | Did we show it examples and let it find the pattern, instead of coding rules by hand? |
| <b>DL</b> | A subset of ML using layered neural networks, strong on images/audio/text    | Is the input a picture, sound, or free-form text rather than a neat spreadsheet?      |

### The Spam Filter, Through Three Eras

**Rule-based AI (2000s):** block any email containing the word "lottery." Easy to trick by misspelling it.

**Machine Learning (2010s):** the model learns from thousands of labelled emails which word combinations and sender patterns usually mean spam – adapts automatically as new tricks appear.

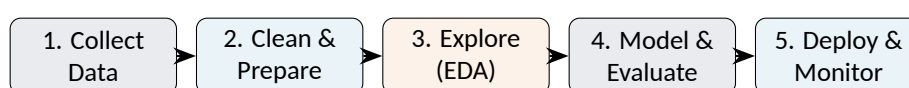
**Deep Learning (today):** the model reads the whole email for context, the way a human would, and catches spam that contains no "obvious" spam words at all.

### Key Takeaway

Every Deep Learning model is a Machine Learning model. Every Machine Learning model is a form of AI. But not every AI system uses learning, and most everyday business problems (loan approval, customer churn, pricing) are solved perfectly well with classical Machine Learning – no Deep Learning required.

## 3 The Data Science Lifecycle

Almost every data science project – including the six mini-projects in this course – follows the same five stages:



- **Collect:** gather raw data – from files, databases, APIs, or web scraping. Real data is often incomplete.
- **Clean & Prepare:** fix inconsistent entries (e.g. "MS Dhoni" vs "M.S. Dhoni"), handle missing values – this is where most of the time actually goes.

- **Explore (EDA):** visualise and summarise the data to find patterns before modelling – what you'll practice heavily in Module 1.
- **Model & Evaluate:** try algorithms (starting simple, e.g. logistic regression, before anything complex) and measure how well they perform.
- **Deploy & Monitor:** put the model into a real application where it has to give fast, reliable answers, and keep checking it still works over time.

## 4 Your 80-Hour Roadmap

| Mod | Title                                       | Hrs | Mini-Project                                                              |
|-----|---------------------------------------------|-----|---------------------------------------------------------------------------|
| 1   | Python for Data Science                     | 5   | EDA on India's COVID-19 dataset (10+ charts, written report)              |
| 2   | Statistics & Probability for ML             | 14  | A/B test analysis for e-commerce pricing strategy                         |
| 3   | Supervised Learning                         | 16  | Loan default prediction – 4 classifiers compared (LR, RF, XGBoost, SVM)   |
| 4   | Unsupervised Learning & Feature Engineering | 16  | Customer segmentation using K-means on RFM features                       |
| 5   | Deep Learning & Neural Networks             | 15  | Plant disease detection with CNN + transfer learning, deployed via Gradio |
| 6   | NLP & Time Series                           | 14  | E-commerce review intelligence: sentiment + NER + ARIMA sales forecast    |

### By the End of This Course

You will have **6 completed, portfolio-ready projects**, each built on a real dataset (IPL, COVID-19, Titanic-style banking data, retail transactions, plant leaf images, and e-commerce reviews). This portfolio is what most recruiters look at first.

## 5 Tools You Will Use

| Category          | Tools                                          |
|-------------------|------------------------------------------------|
| Core Python       | NumPy, Pandas, Matplotlib, Seaborn, Jupyter    |
| Statistics        | scipy, statsmodels                             |
| Classical ML      | scikit-learn, XGBoost, LightGBM                |
| Deep Learning     | TensorFlow, Keras, ResNet, VGG16, LSTM, Gradio |
| NLP & Forecasting | spaCy, NLTK, Word2Vec, ARIMA, FB Prophet       |

## 6 Where This Can Take You

- **Data Analyst** – builds strongly on Modules 1–2 (Pandas, statistics, dashboards).
- **Machine Learning Engineer** – builds strongly on Modules 3–5 (model building, deployment).
- **Data Scientist** – the full stack, Modules 1–6 combined with business judgment.
- **NLP / Computer Vision Specialist** – goes deeper into Modules 5–6.

### Glossary

**Feature** – an individual measurable input used by a model (e.g. distance, time of day).

**Model** – the trained system that turns inputs (features) into a prediction.

**Training** – the process of showing a model examples so it can learn patterns.

**EDA (Exploratory Data Analysis)** – summarising and visualising data before modelling it.

**Label** – the known correct answer for a training example (e.g. “spam” or “not spam”).

**Before Next Session**

1. In your own words, write 2-3 sentences explaining the difference between AI, ML, and DL.
2. Think of one app you use daily. Which stage of the Data Science lifecycle do you think is hardest to get right for that app, and why?
3. Install Python (via Anaconda) and Jupyter Notebook before the next session – we start hands-on with NumPy and Pandas.

*See you in Module 1 – Python for Data Science.*